

First-principles diagrammatic Monte Carlo for electron-phonon interactions and polarons

Yao Luo ¹, Jinsoo Park ^{1,2,3} and Marco Bernardi ¹

¹*Department of Applied Physics and Materials Science, and Department of Physics,
California Institute of Technology, Pasadena, California 91125, USA*

²*Pritzker School of Molecular Engineering,
University of Chicago, Chicago, Illinois 60637, USA*

³*Department of Physics, Pohang University of Science and Technology, Pohang 37673, Korea*

In condensed matter, phonons — quanta of the lattice vibration field— couple with the electrons, leading to the formation of entangled electron-phonon states called polarons. In the intermediate- and strong-coupling regimes common to many conventional and quantum materials, a many-body treatment of polarons requires adding up a large number of electron-phonon Feynman diagrams. In this regard, diagrammatic Monte Carlo is an efficient method for diagram summation and has been employed to study polarons within simplified electron-phonon models. Here we develop diagrammatic Monte Carlo calculations based on accurate first-principles electron-phonon interactions, enabling numerically exact results for ground-state and dynamical properties of polarons in real materials. We implement these calculations in LiF, SrTiO₃, and rutile and anatase TiO₂ and describe both localised and delocalised polarons. Our work enables precise modeling of electron-phonon interactions and polarons in coupling regimes ranging from weak to strong. The results will provide deeper insights into transport phenomena, linear response, and superconductivity within the strong electron-phonon coupling regime.

Polarons are composite quasiparticles consisting of an electron or hole accompanied by a phonon cloud. Because of their fundamental importance in physics^{3,4}, chemistry⁵ and materials science⁶, intense research efforts have focused on polarons starting as early as the 1930s⁷. Polarons are present in systems with strong electron-phonon (*e-ph*) interactions and are crucial to understanding the physics of materials with ionic bonds and/or localized electronic states. In its general form, this strong-coupling problem is common to other quasiparticles interacting with generic bosonic fields, including spin polarons⁸, exciton polarons⁹, and strong light-matter interactions¹⁰.

In the zoo of theoretical methods for polarons, diagrammatic Monte Carlo (DMC) is regarded as a gold standard because it includes *e-ph* diagrams to all orders of perturbation theory. DMC provided numerically exact results for Fröhlich polarons in the entire range of coupling strengths^{1,11}, and its capabilities have been demonstrated in a series of papers^{12–18}. To date, DMC is still limited to model Hamiltonians which cannot provide quantitative predictions of polarons and related properties in real systems.

First-principles calculations based on density functional theory (DFT)¹⁹ and its linear-response variant (DFPT)²⁰ provide accurate *e-ph* Hamiltonians^{21,22}. Leveraging these techniques, several first-principles treatments of polarons have been proposed, including variational wave function approaches for self-trapped polarons^{23,24}, canonical transformation^{25,26}, and diagrammatic methods^{27–29}. Unlike DMC, these methods are approximate, and can mainly address ground state properties, such as the polaron formation energy, or in one case polaron transport in a specific range of coupling strengths²⁸.

In this work, we develop first-principles *e-ph* DMC (FEP-DMC) calculations that can access the spectral and dynamical properties of polarons – including polaron dispersion, effective mass, quasiparticle weight, phonon cloud distribution, spectral function, and mobility – over the entire range of coupling strengths, thereby enabling precise studies of polarons in a wide range of materials. This advance is made possible by overcoming two key challenges.

First, *e-ph* interactions computed from first-principles utilize large multi-dimensional matrices²; this is a major computational bottleneck because FEP-DMC needs to evaluate billions of *e-ph* diagrams, each with up to hundreds of *e-ph* couplings. We overcome this barrier by leveraging our recent technique for data-driven compression of *e-ph* interactions², thereby greatly reducing memory and computational cost. This parsimonious modeling of *e-ph* interactions makes FEP-DMC calculations possible.

Second, the presence of multiple electronic bands causes a sign problem in the e -ph Monte Carlo sampling, which is analogous to the sign problem in fermionic quantum Monte Carlo although it has a different origin³⁰. We address this challenge by deriving a matrix-product formalism to evaluate the amplitude of e -ph Feynman diagrams, where the band indices are summed over exactly using matrix products ordered on the imaginary time axis rather than sampled stochastically. This matrix-product approach greatly alleviates the multi-band sign problem, enabling FEP-DMC calculations in materials with generic electronic band structures and phonon dispersions. With these two barriers removed, our FEP-DMC algorithm can sum all e -ph diagrams with quantitative accuracy. We show its capabilities by studying a range of polaronic materials: LiF, SrTiO₃, and TiO₂ in the rutile and anatase phases.

The polaron is described as a single electron (or hole) interacting with a bath of phonon modes. To leading order in the atomic displacements, the e -ph Hamiltonian reads³¹

$$H = \sum_{n\mathbf{k}} \epsilon_{n\mathbf{k}} \hat{c}_{n\mathbf{k}}^\dagger \hat{c}_{n\mathbf{k}} + \sum_{\mathbf{Q}} \omega_{\mathbf{Q}} \hat{b}_{\mathbf{Q}}^\dagger \hat{b}_{\mathbf{Q}} + \sum_{mn\mathbf{k}, \mathbf{Q}} g_{mn}(\mathbf{k}, \mathbf{Q}) \hat{c}_{m\mathbf{k}+\mathbf{q}}^\dagger \hat{c}_{n\mathbf{k}} (\hat{b}_{\mathbf{Q}} + \hat{b}_{-\mathbf{Q}}^\dagger) \quad (1)$$

where $\epsilon_{n\mathbf{k}}$ are electronic band energies, with band index n and crystal momentum $\mathbf{k}$, the composite index $\mathbf{Q} = (\nu, \mathbf{q})$ labels the phonon mode ν with momentum $\mathbf{q}$, $\omega_{\mathbf{Q}}$ are phonon energies, and $g_{mn}(\mathbf{k}, \mathbf{Q})$ are e -ph coupling matrix elements²¹. The creation and annihilation operators are $\hat{c}_{n\mathbf{k}}^\dagger$ and $\hat{c}_{n\mathbf{k}}$ for electrons, and $\hat{b}_{\mathbf{Q}}^\dagger$ and $\hat{b}_{\mathbf{Q}}$ for phonons.

When the e -ph interactions are sufficiently strong, the electron (or hole) will polarize the surrounding lattice, forming a phonon cloud. The emerging quasiparticle – the polaron – is an entangled state of the electron and phonons. Its wave function can be expressed using many-body theory as

$$|\Psi_0(\mathbf{p})\rangle = \sum_N \sum_{n, \mathbf{Q}_1, \dots, \mathbf{Q}_N} \Psi^{(N)}(\mathbf{p}; n\mathbf{k}, \mathbf{Q}_1, \dots, \mathbf{Q}_N) \times \hat{c}_{n\mathbf{k}}^\dagger \hat{b}_{\mathbf{Q}_1}^\dagger \dots \hat{b}_{\mathbf{Q}_N}^\dagger |0\rangle, \text{ with } \mathbf{p} = \mathbf{k} + \sum_{i=1}^N \mathbf{Q}_i, \quad (2)$$

where N is phonon number and $|0\rangle$ is the vacuum state. The total (center-of-mass) momentum of the polaron, $\mathbf{p}$, is the sum of the electron and phonon momenta and is a good quantum number to label the polaron state. The energy minimum at each momentum $\mathbf{p}$ gives the polaron band dispersion, $E_0(\mathbf{p})$, whose minimum gives the polaron ground state

energy, $E_0(\mathbf{p}_{\text{GS}})$. The polaron effective mass, m_p , can be extracted from $E_0(\mathbf{p})$ along with the mass enhancement factor m_p/m_e (where m_e is the electron effective mass), which is a measure of polaron localization.

The tensor $\Psi^{(N)}(\mathbf{p}; n\mathbf{k}, \mathbf{Q}_1, \dots, \mathbf{Q}_N)$ in Eq. (2) fully characterizes the polaron state. In particular, we define the N -phonon amplitude

$$Z^{N,\mathbf{p}} = \sum_{n, \mathbf{Q}_1, \dots, \mathbf{Q}_N} |\Psi^{(N)}(\mathbf{p}; n\mathbf{k}, \mathbf{Q}_1, \dots, \mathbf{Q}_N)|^2, \quad (3)$$

which can be interpreted as the phonon number distribution in the polaron cloud for polaron state $|\Psi_0(\mathbf{p})\rangle$, with the sum rule $\sum_N Z^{N,\mathbf{p}} = 1$. This allows us to calculate the average number of phonons in the polaron cloud, $\langle N_{\text{ph}} \rangle = \sum_N N Z^{N,\mathbf{p}}$, and resolve the mode contributions. The quantities $Z^{N,\mathbf{p}}$ and $\langle N_{\text{ph}} \rangle$ characterize the phonon cloud and lattice distortion induced by the charge carrier. For weak coupling, the polaron state is band-like, with $m_p/m_e \approx 1$ and a negligible phonon cloud $\langle N_{\text{ph}} \rangle \ll 1$ surrounding the charge carrier. For strong coupling, the polaron state is dressed by a large number of phonons, $\langle N_{\text{ph}} \rangle \gg 1$, and tends to localize due to its heavier mass $m_p/m_e \gg 1$. DMC works equally as well in both limits.

Similarly, we compute the contribution of each electronic state $|n\mathbf{k}\rangle$ to the polaron state $|\Psi_0(\mathbf{p})\rangle$ from the electronic reduced density matrix $\hat{\rho}_e^{\mathbf{p}}$, obtained by tracing out the phonons:

$$\hat{\rho}_e^{\mathbf{p}} = \text{Tr}_{\text{ph}} (|\Psi_0(\mathbf{p})\rangle \langle \Psi_0(\mathbf{p})|) = \sum_{mn\mathbf{k}} \rho_e^{\mathbf{p}}(mn\mathbf{k}) |m\mathbf{k}\rangle \langle n\mathbf{k}|.$$

Its diagonal matrix elements $f_e^{\mathbf{p}}(n\mathbf{k}) = \langle n\mathbf{k} | \hat{\rho}_e^{\mathbf{p}} | n\mathbf{k} \rangle$ take the general form

$$f_e^{\mathbf{p}}(n\mathbf{k}) = Z^{0,\mathbf{p}}(n) \delta(\mathbf{p} - \mathbf{k}) + \tilde{f}_e(n\mathbf{k}), \quad (4)$$

consisting of a Dirac delta function with weight $Z^{0,\mathbf{k}}(n)$ plus a continuum part $\tilde{f}_e(n\mathbf{k})$ quantifying the electronic contributions to the polaron (Supplementary Section I).

The quantities defined above provide a comprehensive description of polaron properties. Each quantity can be expanded as a Taylor series in the e -ph interactions and written as a sum of e -ph Feynman diagrams:

$$Q(y) = \sum_{s=0}^{\infty} \sum_{\mathbf{T}_s} \int dx_1 \dots dx_s \mathcal{D}(y; \mathbf{T}_s, x_1, \dots, x_s), \quad (5)$$

where s is the expansion order, $\mathbf{T}_s$ loops over the topologically distinct diagrams at order s , and x_i are internal variables. DMC evaluates this high-dimensional integral using Markov

chain Monte Carlo (MC) in the space of e -ph Feynman diagrams^{11,32}. The theory and numerical evaluation of physical quantities in DMC are described in Supplementary Section I.

We demonstrate FEP-DMC calculations in several polaronic materials – LiF, SrTiO₃, and the anatase and rutile TiO₂ phases. We obtain their electronic structure and phonon dispersions using the QUANTUM ESPRESSO package³³, and employ PERTURBO²¹ in conjunction with WANNIER90³⁴ to compute and interpolate the e -ph interactions, which are used in compressed form, $g_{mn}(\mathbf{k}, \mathbf{Q}) \approx \sum_{\gamma} S_{mn}^{\gamma} u_{mn}^{\gamma}(\mathbf{k}) v_{mn}^{\gamma}(\mathbf{Q})$, using the Luo-Bernardi (LB) compression technique². Additional computational details are provided in Methods, while convergence tests and error analysis are available in Supplementary Section II.

FEP-DMC samples Feynman diagrams according to the amplitude $\mathcal{D}$ in Eq. (5) and the sign problem arises because $\mathcal{D}$ is negative for some diagrams. Our implementation relies on a matrix-product approach to alleviate (in fact, nearly remove) the sign problem from multiple electronic bands (Supplementary Section I). In Fig. 1(a), we compute the electron Green’s function for a hole polaron in LiF and compare our matrix-product approach with explicit band sampling. At small imaginary times τ , the two methods give consistent results, but for larger values of τ the statistical error for explicit band sampling (see the shaded orange region in Fig. 1(a)) grows exponentially faster than in the matrix-product approach. This behavior can be understood from the average sign $S(\tau)$ shown in Fig. 1(b), defined as the mean value of the sign of the Green’s function diagrams with fixed external time τ (see Eq. (S25) in Supplementary Section I). The average sign drops below 10^{-3} at short times in the band sampling approach, causing a rapid growth of MC error in Fig. 1(a); the matrix-product method also has an exponentially decaying sign as a function of time, but with a much slower rate. The apparent sign problem in band sampling is mainly a result of inter-band coherence in the first-principles e -ph Hamiltonian when multiple electronic bands are present. The matrix-product calculation sums over all band configurations, leading to a sign cancellation effect. The small residual sign problem in the matrix-product approach originates from the intrinsic $\mathbf{k}$ -dependence of the e -ph matrix elements, as exemplified by the SSH polaron model, and could be addressed in future work using bold-line DMC¹⁴.

We first apply our approach to calculations of polaron ground state properties. Figure 2 shows the polaron formation energy ΔE_0 , defined as the polaron ground state energy relative to the DFT electronic band extrema, in LiF (both electron and hole carriers), SrTiO₃, and anatase TiO₂. In all these systems, the ground state energy is found for polaron mo-

momentum $\mathbf{p}_{GS} = 0$ (Γ -point in Fig. 2). These calculations employ uniform grids with $N_{k_i}^3$ $\mathbf{k}$ -points, which corresponds to a solid with $N_{k_i}^3$ unit cells. Our LB compression technique² allows us to reach grid sizes of up to 50^3 (for TiO_2) and 120^3 $\mathbf{k}$ -points for LiF and SrTiO_3 , enabling accurate extrapolation of the formation energy to the thermodynamic limit (TDL) for both small (localized) and large (delocalized) polarons. (Convergence of ΔE_0 with respect to hyperparameters such as the maximum imaginary time and number of singular values is discussed in Supplementary Sections II A and II B.) We have verified that our implementation is accurate by reproducing results for the Holstein³⁵ and Fröhlich models¹⁷ (see Supplementary Section II. K).

The slow convergence of the formation energy with respect to the grid size in Fig. 2(b)-(d) indicates the presence of large polarons extending over many unit cells in LiF, SrTiO_3 and anatase TiO_2 . Conversely, the formation energy for hole polaron in LiF, shown in Fig. 2(a), converges rapidly, reflecting the small hole-polaron size in LiF. Corrections to the polaron energy from electronic bands with higher energy than those included in the FEP-DMC calculation are obtained using the Allen-Heine-Cardona theory, including both Fan-Migdal and Debye-Waller terms,^{36,37} and added to our FEP-DMC results. We carry out these calculations for electrons and holes in LiF and obtain corrected formation energies of $\Delta E_0 = -0.538$ eV for electrons and $\Delta E_0 = 2.270$ eV for holes in LiF, consistent with previous results²⁹ (see details in Supplementary Section II G.)

FEP-DMC provides the exact polaron wave function in the many-particle basis, as defined in Eq. (S3). This allows us to access dynamical and spectral properties of polarons. Figure 3 illustrates the electron and phonon contributions to the polaron wave function in LiF, SrTiO_3 and anatase TiO_2 . The upper panel of Fig. 3 visualizes the continuum part $\tilde{f}_e(n\mathbf{k})$, defined in Eq. (4), for the polaron ground state in these systems. For cases where electrons form large polarons, $\tilde{f}_e(n\mathbf{k})$ is localized around $\mathbf{k} = \mathbf{p}$ (here, the Γ point), but when a small polaron forms, such as for holes in LiF, $\tilde{f}_e(n\mathbf{k})$ spreads over the entire Brillouin zone. Therefore, $\tilde{f}_e(n\mathbf{k})$ provides a fingerprint for the character of the polaron state.

Analyzing the phonon sector of the polaron ground state sheds light on the size and nature of its dynamical phonon cloud. The lower panels in Fig. 3 show the N -phonon amplitude $Z^{N,\mathbf{p}=\Gamma}$ defined above in Eq. (3). For the small hole polaron in LiF, the N -ph amplitude follows a Poisson distribution, peaked at an average phonon number of $\langle N_{\text{ph}} \rangle \approx 26$, consistent with Lang-Firsov transformation results for small polarons^{38,39}. This means that a single

hole in LiF is dressed by 26 phonons on average; this dense phonon cloud is made up entirely by longitudinal optical (LO) phonons with both long-range and strong local e-ph interactions (see the inset in Fig. 3(a)). We have verified that the short- and long-range interactions give approximately equal contributions to the hole small-polaron formation energy in LiF.

For the other materials studied in Fig. 3, the zero-phonon quasiparticle weight $Z(N=0)$ is finite, and the N -phonon distribution is skewed toward small $\langle N_{\text{ph}} \rangle$ values between 1 – 4, consistent with the presence of large polarons. In all cases, the LO phonons entirely govern the large-polaron formation.

Our approach also enables direct calculations of polaron dispersion curves, for both small and large polarons (see Fig. 4). In all the materials we studied, the polaron band is significantly narrower than the electronic band. The corresponding mass enhancements, obtained with a parabolic fit, are given in Table I for LiF, SrTiO₃ and anatase TiO₂ together with other polaron results for these systems. For small hole polarons in LiF, the polaron dispersion is too narrow to extract the effective mass – the polaron bandwidth is less than 0.04 eV, nearly two orders of magnitudes smaller than the electronic bandwidth (Fig. 4(a)). This result signals that the hole polaron in LiF is self-trapped. For electron polarons in LiF, SrTiO₃, and anatase TiO₂ the polaron bands are dispersive, showing the presence of large polarons in these materials, consistent with experiments^{40–42} (Fig. 4(b)-(d)). For anatase TiO₂, the mass enhancement is 1.6 along G-X, in agreement with the value of 1.7 extracted from ARPES measurements in Ref.⁴⁰ and a previous cumulant study⁴³. For LiF, ARPES data is not available because the material is difficult to dope. FEP-DMC can correctly describe the polaritonic regime where the polaron band is degenerate with strongly-coupled LO phonons, leading to the emergence of flat polaron bands at finite momenta with mixed polaron and phonon characters. We find this trend in anatase TiO₂ (see Fig. 4(d)), where the polaron dispersion becomes flat when the energy exceeds the one-phonon continuum at ~ 100 meV above the polaron band minimum⁴⁴. Finally, we employ FEP-DMC to study charge transport and compute the polaron mobility in the small- and large-polaron regimes; this advance builds on work by Mishchenko et al. on Holstein and Fröhlich polaron models^{13,16,18}. Briefly, we compute the current-current correlation function in imaginary frequency using FEP-DMC (Supplementary Section I C),

$$\chi_n^{\alpha\alpha'} = k_B T \int_0^\beta d\tau \int_0^\beta d\tau' e^{i\omega_n(\tau-\tau')} \langle \hat{T} [\hat{J}^\alpha(\tau) \hat{J}^{\alpha'}(\tau')] \rangle,$$

where $\hat{T}$ is the time-ordering operator, $\hat{J}^\alpha = -e \sum_{n\mathbf{k}} v_{n\mathbf{k}}^\alpha \hat{c}_{n\mathbf{k}}^\dagger \hat{c}_{n\mathbf{k}}$ is the current operator, $\alpha \in (x, y, z)$ are Cartesian directions, $\beta = 1/k_B T$, and $v_{n\mathbf{k}}^\alpha$ is the velocity of the electronic state $|n\mathbf{k}\rangle$ in the α -direction. We then extract the optical conductivity from the current-current correlation function using analytic continuation with the stochastic optimization method (SOM)^{11,45}, and obtain the dc mobility from the conductivity in the zero-frequency limit. The ill-defined numerical analytic continuation can be formulated as a well-posed question using SOM with consistent constraints⁴⁵. We remark that our FEP-DMC, when applied to single-band model e-ph Hamiltonians, gives the same results as previous e-ph DMC methods. Therefore, FEP-DMC can describe both the small and large polaron transport regimes as reported in previous DMC studies on the Frohlich and Holstein models^{13,16,18}. To show this point, we reproduce the temperature-dependent mobility for small polarons in the Holstein model and large polarons in the Fröhlich model with strong e-ph coupling (see Supplementary Section II. K).

Focusing on electron polarons in rutile and anatase TiO_2 (Fig. 5(a)), we use FEP-DMC to compute the mobility as a function of temperature in rutile TiO_2 (Fig. 5(b)-(c)) and anatase TiO_2 (Fig. 5(d)-(e)), and compare our results with available experimental data. In rutile, the predicted mobility along the a crystallographic axis exhibits small-polaron transport with three distinct temperature regimes. At low temperatures ($T < 300$ K), the mobility decreases with temperature showing a metallic-like behavior. At intermediate temperatures between 300 – 500 K, the mobility increases with temperature, consistent with the experimental observation of thermally activated charge transport in rutile⁴⁶ and recent molecular dynamics simulations⁴⁷ (dashed line in Fig. 5(b)). At high temperatures ($T > 500$ K), the mobility nearly saturates and in fact slightly decreases with temperature. This non-monotonic temperature-dependent mobility is characteristic of small-polaron transport, as reported in previous DMC¹⁶ and dynamical mean-field theory⁴⁸ studies. Measurements from Ref.⁴⁹ only capture the low-temperature metallic behavior but do not observe thermally activated transport at higher temperatures, which may be due to Ti-interstitial doping. The predicted c -axis mobility, μ_c , shown in Fig. 5(c), is in excellent agreement with the experimental data. In our results, μ_c saturates at $T > 400$ K, consistent with polaronic transport for intermediate-to-strong e-ph coupling regime. The different behavior for the mobility in the a - and c -axis crystallographic directions occurs because the lowest conduction band is significantly less dispersive in the a and b directions compared to the c direction. Future

work will aim to compare our results with transport measurements in purer high-quality rutile TiO_2 samples. Studies of polaron hopping typically rely on simulations using large supercells with an extra charge added explicitly⁶. FEP-DMC can describe this charge hopping regime using only unit cell calculations and first-principles e -ph interactions, providing quantitatively accurate mobility predictions.

The computed mobility is similarly accurate in anatase TiO_2 , as shown in Fig. 5(d)-(e), where the FEP-DMC results are in excellent agreement with experiments^{50,51}, both in terms of temperature dependence and absolute value. For comparison, using only lowest-order e -ph interactions plus the Boltzmann equation (BTE curve in Fig. 5(d)) leads to a greatly overestimated mobility, which is also observed in another paper⁵². The BTE overestimating the mobility is unrelated to the specific settings employed in the first-principles calculations. Rather, it is a consequence of the BTE's inability to capture the many-body effects deriving from strong e -ph interactions and quasiparticle renormalization in TiO_2 , as shown in the spectral function in Fig. 5(f) (see Supplementary Section II. H for details of BTE calculations.) For both the a - and c -directions, the temperature dependence of the mobility from FEP-DMC follows a power law, indicating transport of mobile large polarons in anatase. Such power-law mobility has been observed in several oxides with large polarons, including SrTiO_3 ^{28,53}. Our FEP-DMC calculations for the two phases of TiO_2 are consistent with the experimental observation that excess charge carriers behave as small polarons in rutile TiO_2 and large polarons in anatase TiO_2 ⁵⁴. Our FEP-DMC results for charge transport in SrTiO_3 are also in very good agreement with experiments (see Supplementary Section II.J).

The spectral function can be extracted from the FEP-DMC imaginary-time Green's function using analytic continuation⁵⁵ (see details in Supplementary Section II. E.). Figure 5(f) shows the computed spectral function for the electronic state at the conduction band minimum in anatase TiO_2 , and compares it with that obtained from the cumulant expansion (CE) method^{27,28,56}. The CE spectral function consists of a quasiparticle peak with reduced weight and multiple satellites at higher energy. These satellites, also known as phonon sidebands, are a hallmark of polaronic behavior — they have been observed in experiments in TiO_2 ⁴⁰ and both predicted and observed in other polar oxides^{28,57}. The FEP-DMC spectral function agrees with the CE result near $\mathbf{k} = \Gamma$. Yet, the quasiparticle weight $Z_{\mathbf{k}}^0$ from CE gradually deviates from the FEP-DMC results as shown in the inset of Fig. 5(f). The limitation of the CE at finite momentum, which is known from Holstein model results⁵⁸,

is overcome by our FEP-DMC calculations, which provide accurate spectral properties at arbitrary momenta. At present, the need for analytic continuation in FEP-DMC limits the accuracy of satellites carrying a small spectral weight.

DISCUSSION

Although in DMC the polaron wave function is translationally invariant, one can still extract the lattice distortion associated with electron or hole polarons. We define the e -ph distortion function, $d(\mathbf{r}_{ep}) = \sum_{\mathbf{r}_e} \langle \Psi_0 | \hat{c}_{\mathbf{r}_e}^\dagger \hat{c}_{\mathbf{r}_e} \hat{\mathbf{x}}_{\mathbf{r}_e + \mathbf{r}_{ep}} | \Psi_0 \rangle$, which quantifies how an atomic displacement $\mathbf{x}$ at position $\mathbf{r}_{ep}$ relative to the electron position $\mathbf{r}_e$ couples with the electronic density at $\mathbf{r}_e$. This e -ph distortion function is compatible with a translationally-invariant polaron wave function $|\Psi_0\rangle$ and can quantify the polaron-induced atomic displacements in real space. We plan to study this quantity in future work using FEP-DMC.

Our mobility calculations employ the standard current operator used in band theory. In future work, the current operator could be extended to include a phonon-assisted term, which is proportional to the strength of inter-site e -ph coupling⁵⁹ and is important in some systems, particularly in organic crystals with strong nonlocal e -ph coupling⁶⁰. For the materials studied here, we have verified that the inter-site e -ph interactions are far weaker than the intra-site e -ph interactions, and give only a small contribution (10–20% of the total) to the Fan-Migdal e -ph self-energy (see Fig. S9 in Supplementary Section II.I). This implies that the phonon-assisted correction to the mobility can be neglected in this work. We plan to explore phonon-assisted transport within FEP-DMC in future work.

In summary, we develop first-principles DMC calculations describing e -ph interactions of arbitrary strengths with quantitative accuracy, setting a gold standard for modeling polarons and e -ph physics in materials. This advance paves the way for studies of the strong e -ph coupling regime in conventional and quantum materials, with applications to electronics, energy, low-dimensional materials, quantum systems, and spectroscopy. The method can also be extended to treat strong coupling between phonons and other excitations – spin, excitons, magnons, etc. – pushing first-principles calculations in condensed matter beyond the weak-coupling regime. We plan to explore these exciting directions in future work.

ACKNOWLEDGEMENTS

Y.L. thanks Ivan Maliyov, Junjie Yang and Alice Lee for fruitful discussions. Y. L. acknowledges partial support from the Eddleman Graduate Fellowship. M.B. is grateful to the Scuola Normale Superiore in Pisa, Italy, for hosting him during the writing of this manuscript. Method development was supported by the U.S. Department of Energy, Office of Science, Office of Advanced Scientific Computing Research and Office of Basic Energy Sciences, Scientific Discovery through Advanced Computing (SciDAC) program under Award No. DESC0022088. Code development was supported by the National Science Foundation under Grant No. OAC-2209262. Calculations of transport and polarons in oxides were supported by the AFOSR and Clarkson Aerospace Corp under award FA9550-24-1-0004. This research used resources of the National Energy Research Scientific Computing Center (NERSC), a U.S. Department of Energy Office of Science User Facility using NERSC award DDR-ERCAP0026831.

AUTHOR CONTRIBUTIONS

Y.L. and M.B. conceived and designed the research. Y.L. performed the calculations and analysis with support from all authors. Y.L. and M.B. wrote the manuscript with input from all authors.

COMPETING INTERESTS

The authors declare no competing interests.

TABLES

	LiF(h)	LiF(e)	SrTiO ₃ (e)	Anatase(e)
ΔE_0 (eV)	2.26	-0.408	-0.181	-0.178
$\langle N_{\text{ph}} \rangle$	26.5	4.22	1.15	0.65
m_{p}/m_e	NA	6.1 ($\Gamma - L$)	1.4 ($\Gamma - M$)	1.2 ($\Gamma - Z$)

TABLE I. Computed polaron properties in LiF, SrTiO₃ and anatase TiO₂, including the formation energy ΔE_0 , the average phonon number $\langle N_{\text{ph}} \rangle$, and the mass enhancement factor m_{p}/m_e along the directions given in parentheses.

FIGURE LEGENDS/CAPTIONS

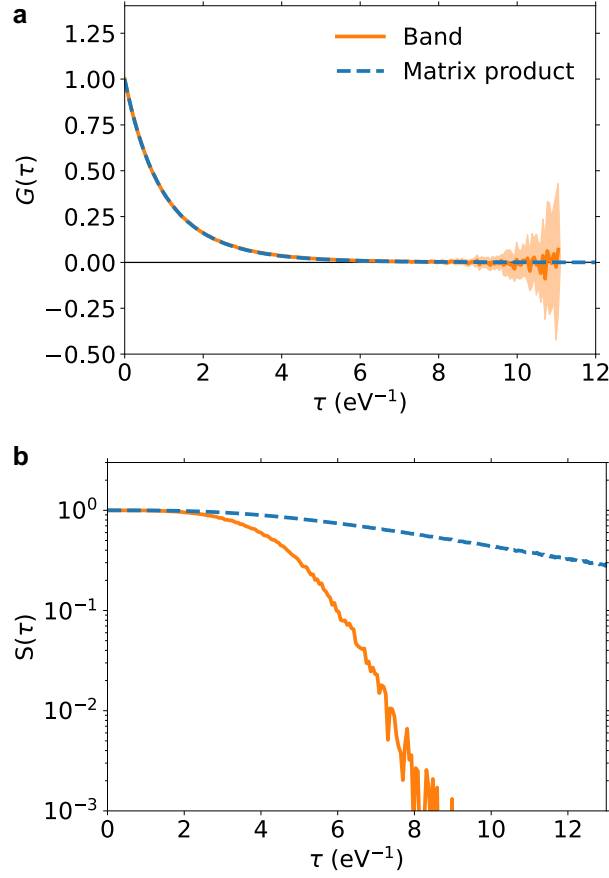


Figure 1. Sign problem in e -ph DMC. (a) Imaginary time electronic Green's function for hole polaron in LiF from the matrix-product and band sampling methods. The shaded orange region shows the standard error of the mean (SEM) evaluated from 20 independent MC simulations. Both methods use the same number of MC updates. (b) The average sign of the Green's function diagram as a function of the external time τ . Note that an average sign of $S = 1$ would ideally remove the sign problem.

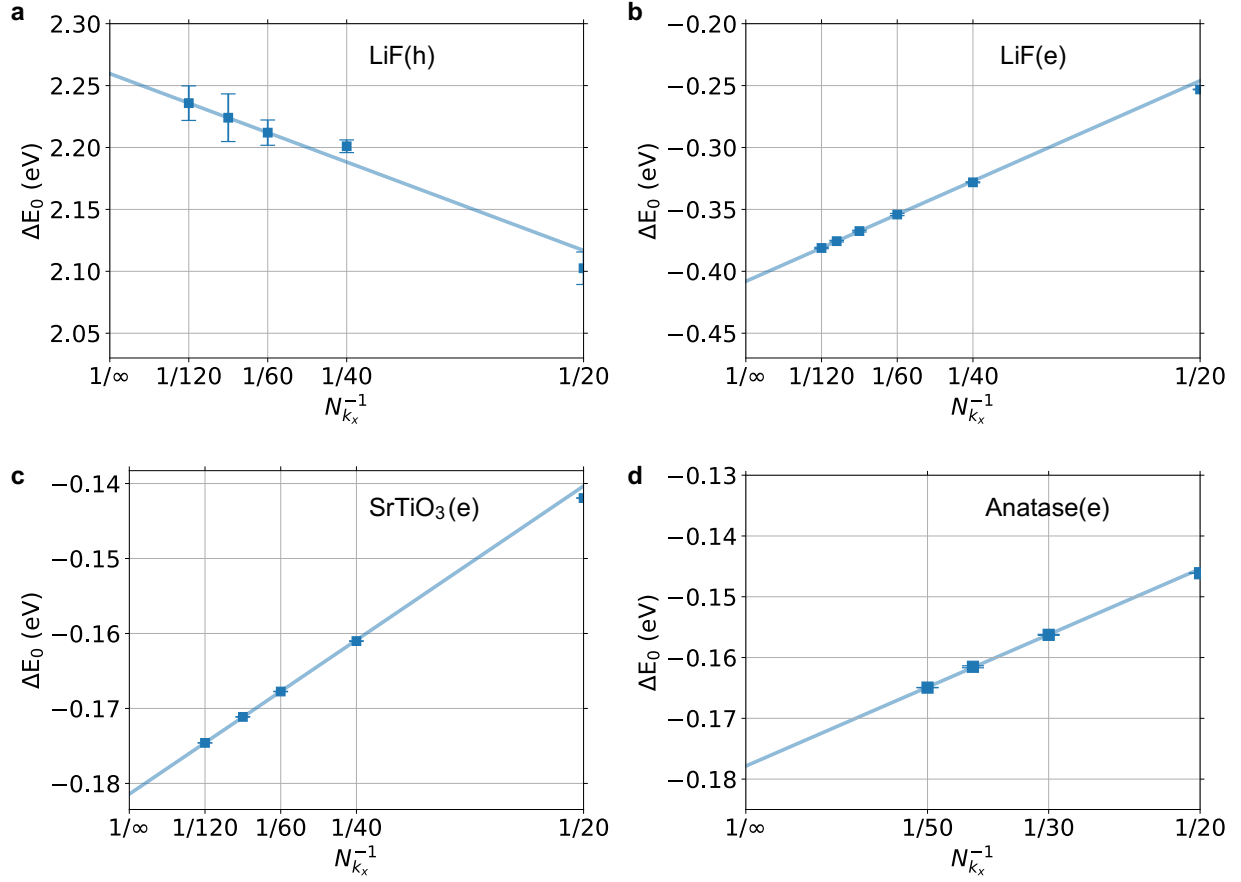


Figure 2. Polaron formation energy, shown as a function of inverse grid size for (a) hole polaron in LiF, and electron polarons in (b) LiF, (c) SrTiO₃ and (d) anatase TiO₂. We use data from the three largest grid sizes to extrapolate the polaron formation energy to the TDL. The 'e' and 'h' symbols in parentheses indicate electron and hole polarons respectively. Data are presented as mean values with error bars equal to $\pm$ SEM, where the SEM is extracted from 4–10 independent MC simulations.

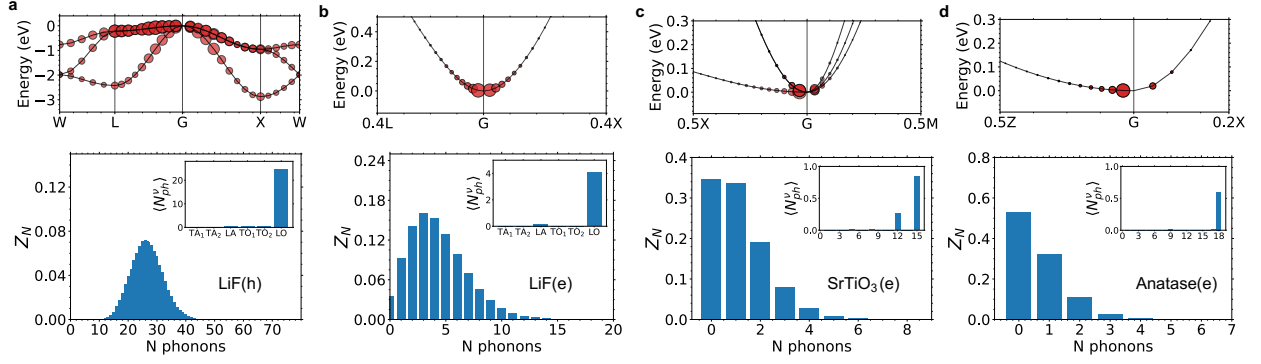


Figure 3. Analysis of the polaron ground state, showing both the electron (top) and phonon (bottom) contributions. Results are shown for (a) hole polaron in LiF, and electron polarons in (b) LiF, (c) SrTiO₃ and (d) anatase TiO₂. The top panels show the continuum part of the diagonal reduced density matrix, $\tilde{f}_e(n\mathbf{k})$ in Eq. (4), with dot sizes proportional to $\tilde{f}_e^{\mathbf{p}=\Gamma}(n\mathbf{k})$. The bottom panels show the N -phonon amplitude $Z^{N,\mathbf{p}=\Gamma}$, with insets showing the average phonon number separately for each phonon mode.

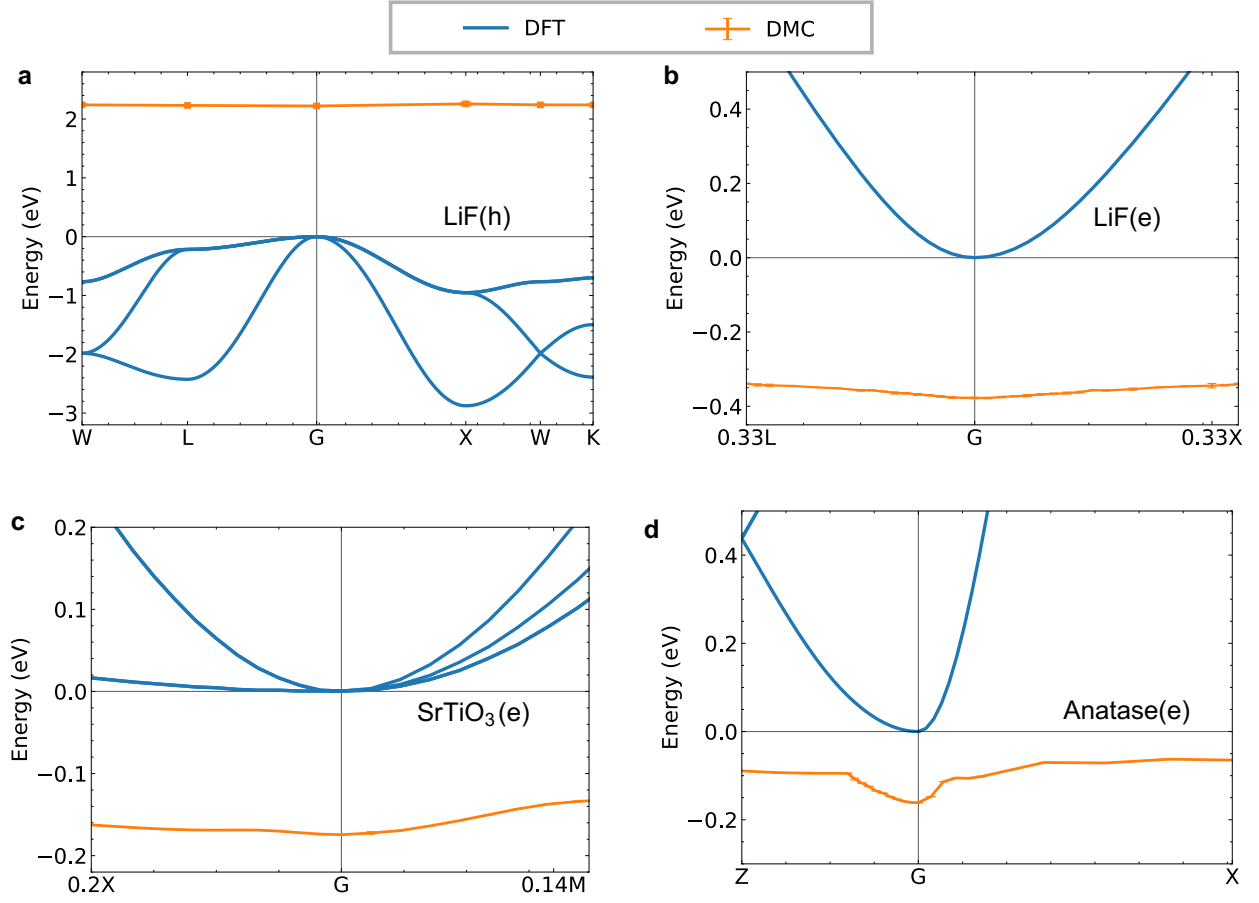


Figure 4. Polaron dispersions along high-symmetry paths for (a) LiF hole, (b) LiF electron, (c) SrTiO₃ electron, and (d) anatase TiO₂ electron. The DFT electronic band structure is shown for comparison in each case. The polaron calculations use grids ranging from 50^3 to 120^3 k -points. The error bars are SEM values (here, as small as the linewidth) extracted from 4 – 10 independent MC simulations.

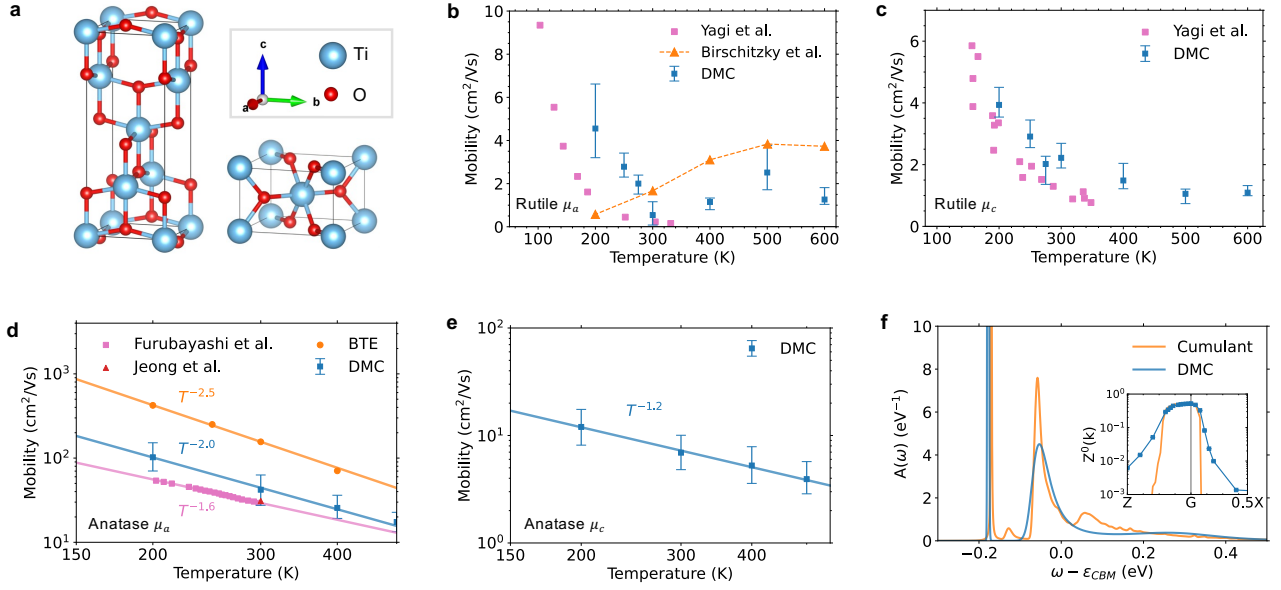


Figure 5. Polaronic charge transport in rutile and anatase TiO₂. (a) The crystal structure of anatase and rutile. (b)-(c) Computed mobility versus temperature for rutile in the *a* and *c* crystallographic directions, respectively. Also shown are molecular dynamics simulation results from Ref.⁴⁷ (dashed line) and experimental data from Ref.⁴⁹. (d)-(e) Mobility in anatase in the *a*- and *c*-directions, respectively. The dots are experimental data for Co-doped⁵¹ and 1% Nb-doped anatase TiO₂⁵⁰. (f) Spectral function for the band-edge electronic state in anatase, comparing results from FEP-DMC and the cumulant method²⁸; the inset shows the quasiparticle weight at finite momenta. The error bars in (b)-(e) are SEM values extracted from 500 independent SOM analytical continuation calculations.

REFERENCES

1. N. V. Prokof'ev and B. V. Svistunov, Polaron problem by diagrammatic quantum Monte Carlo, *Phys. Rev. Lett.* **81**, 2514 (1998).
2. Y. Luo, D. Desai, B. K. Chang, J. Park, and M. Bernardi, Data-driven compression of electron-phonon interactions, *Phys. Rev. X* **14**, 021023 (2024).
3. H. Fröhlich, Electrons in lattice fields, *Adv. Phys.* **3**, 325 (1954).
4. T. Holstein, Studies of polaron motion: Part ii. the “small” polaron, *Ann. Phys. (N. Y.)* **8**, 343 (1959).
5. J. L. Bredas and G. B. Street, Polarons, bipolarons, and solitons in conducting polymers, *Acc.*

- [Chem. Res. **18**, 309 \(1985\).](#)
6. C. Franchini, M. Reticcioli, M. Setvin, and U. Diebold, Polarons in materials, [Nat. Rev. Mater. **6**, 560 \(2021\).](#)
 7. L. D. Landau, Motion of electrons in crystal lattice, *Phys. Z. Sowjetunion* **3**, 664 (1933).
 8. G. Martinez and P. Horsch, Spin polarons in the t-J model, [Phys. Rev. B **44**, 317 \(1991\).](#)
 9. O. Verzele, R. Ferreira, and G. Bastard, Excitonic polarons in semiconductor quantum dots, [Phys. Rev. Lett. **88**, 146803 \(2002\).](#)
 10. J. Bloch, A. Cavalleri, V. Galitski, M. Hafezi, and A. Rubio, Strongly correlated electron–photon systems, [Nature **606**, 41 \(2022\).](#)
 11. A. S. Mishchenko, N. V. Prokof’ev, A. Sakamoto, and B. V. Svistunov, Diagrammatic quantum Monte Carlo study of the Fröhlich polaron, [Phys. Rev. B **62**, 6317 \(2000\).](#)
 12. A. S. Mishchenko and N. Nagaosa, Quasidegenerate self-trapping in one-dimensional charge transfer exciton, [Phys. Rev. Lett. **86**, 4624 \(2001\).](#)
 13. A. S. Mishchenko, N. Nagaosa, N. V. Prokof’ev, A. Sakamoto, and B. V. Svistunov, Optical conductivity of the Fröhlich polaron, [Phys. Rev. Lett. **91**, 236401 \(2003\).](#)
 14. D. J. J. Marchand, G. De Filippis, V. Cataudella, M. Berciu, N. Nagaosa, N. V. Prokof’ev, A. S. Mishchenko, and P. C. E. Stamp, Sharp transition for single polarons in the one-dimensional Su-Schrieffer-Heeger model, [Phys. Rev. Lett. **105**, 266605 \(2010\).](#)
 15. A. S. Mishchenko, N. Nagaosa, and N. Prokof’ev, Diagrammatic Monte Carlo method for many-polaron problems, [Phys. Rev. Lett. **113**, 166402 \(2014\).](#)
 16. A. S. Mishchenko, N. Nagaosa, G. De Filippis, A. de Candia, and V. Cataudella, Mobility of Holstein polaron at finite temperature: An unbiased approach, [Phys. Rev. Lett. **114**, 146401 \(2015\).](#)
 17. T. Hahn, S. Klimin, J. Tempere, J. T. Devreese, and C. Franchini, Diagrammatic Monte Carlo study of Fröhlich polaron dispersion in two and three dimensions, [Phys. Rev. B **97**, 134305 \(2018\).](#)
 18. A. S. Mishchenko, L. Pollet, N. V. Prokof’ev, A. Kumar, D. L. Maslov, and N. Nagaosa, Polaron mobility in the “beyond quasiparticles” regime, [Phys. Rev. Lett. **123**, 076601 \(2019\).](#)
 19. R. M. Martin, *Electronic Structure: Basic Theory and Practical Methods* (Cambridge University Press, 2004).
 20. S. Baroni, S. de Gironcoli, A. Dal Corso, and P. Giannozzi, Phonons and related crystal

- properties from density-functional perturbation theory, [Rev. Mod. Phys. **73**, 515 \(2001\)](#).
21. J.-J. Zhou, J. Park, I.-T. Lu, I. Maliyov, X. Tong, and M. Bernardi, PERTURBO: A software package for ab initio electron-phonon interactions, charge transport and ultrafast dynamics, [Comput. Phys. Commun. **264**, 107970 \(2021\)](#).
 22. L. A. Agapito and M. Bernardi, Ab initio electron-phonon interactions using atomic orbital wave functions, [Phys. Rev. B **97**, 235146 \(2018\)](#).
 23. S. Kokott, S. V. Levchenko, P. Rinke, and M. Scheffler, First-principles supercell calculations of small polarons with proper account for long-range polarization effects, [New J. Phys. **20**, 033023 \(2018\)](#).
 24. M. Reticcioli, U. Diebold, G. Kresse, and C. Franchini, Small Polarons in Transition Metal Oxides, in [Handbook of Materials Modeling](#) (Springer International Publishing, 2020) pp. 1–39.
 25. N.-E. Lee, H.-Y. Chen, J.-J. Zhou, and M. Bernardi, Facile ab initio approach for self-localized polarons from canonical transformations, [Phys. Rev. Mater. **5**, 063805 \(2021\)](#).
 26. Y. Luo, B. K. Chang, and M. Bernardi, Comparison of the canonical transformation and energy functional formalisms for ab initio calculations of self-localized polarons, [Phys. Rev. B **105**, 155132 \(2022\)](#).
 27. J. P. Nery, P. B. Allen, G. Antonius, L. Reining, A. Miglio, and X. Gonze, Quasiparticles and phonon satellites in spectral functions of semiconductors and insulators: Cumulants applied to the full first-principles theory and the Fröhlich polaron, [Phys. Rev. B **97**, 115145 \(2018\)](#).
 28. J.-J. Zhou and M. Bernardi, Predicting charge transport in the presence of polarons: The beyond-quasiparticle regime in SrTiO₃, [Phys. Rev. Res. **1**, 033138 \(2019\)](#).
 29. J. Lafuente-Bartolome, C. Lian, W. H. Sio, I. G. Gurtubay, A. Eiguren, and F. Giustino, Unified approach to polarons and phonon-induced band structure renormalization, [Phys. Rev. Lett. **129**, 076402 \(2022\)](#).
 30. A. J. Kim, P. Werner, and R. Valentí, Alleviating the sign problem in quantum Monte Carlo simulations of spin-orbit-coupled multiorbital Hubbard models, [Phys. Rev. B **101**, 045108 \(2020\)](#).
 31. M. Bernardi, First-principles dynamics of electrons and phonons, [Eur. Phys. J. B **89**, 239 \(2016\)](#).
 32. J. Greitemann and L. Pollet, Lecture notes on Diagrammatic Monte Carlo for the Fröhlich polaron, [SciPost Phys. Lect. Notes , 2 \(2018\)](#).

33. P. Giannozzi *et al.*, QUANTUM ESPRESSO: a modular and open-source software project for quantum simulations of materials, [J. Phys. Condens. Matter **21**, 395502 \(2009\)](#).
34. G. Pizzi, V. Vitale, R. Arita, S. Blügel, F. Freimuth, G. Géranton, M. Gibertini, D. Gresch, C. Johnson, T. Koretsune, J. Ibañez-Azpiroz, H. Lee, J.-M. Lihm, D. Marchand, A. Marrazzo, Y. Mokrousov, J. I. Mustafa, Y. Nohara, Y. Nomura, L. Paulatto, S. Poncé, T. Ponweiser, J. Qiao, F. Thöle, S. S. Tsirkin, M. Wierzbowska, N. Marzari, D. Vanderbilt, I. Souza, A. A. Mostofi, and J. R. Yates, Wannier90 as a community code: new features and applications, [J. Phys. Condens. Matter **32**, 165902 \(2020\)](#).
35. C. Franchini and S. Ragni, *Diagrammatic Monte Carlo study of the Holstein polaron*, Master's thesis.
36. P. B. Allen and M. Cardona, Theory of the temperature dependence of the direct gap of germanium, [Phys. Rev. B **23**, 1495 \(1981\)](#).
37. J.-M. Lihm and C.-H. Park, Phonon-induced renormalization of electron wave functions, [Phys. Rev. B **101**, 121102 \(2020\)](#).
38. S. Ciuchi, F. de Pasquale, S. Fratini, and D. Feinberg, Dynamical mean-field theory of the small polaron, [Phys. Rev. B **56**, 4494 \(1997\)](#).
39. G. D. Mahan, *Many-Particle Physics*, 3rd ed. (Springer, 2000).
40. S. Moser, L. Moreschini, J. Jaćimović, O. S. Barišić, H. Berger, A. Magrez, Y. J. Chang, K. S. Kim, A. Bostwick, E. Rotenberg, L. Forró, and M. Grioni, Tunable polaronic conduction in anatase TiO₂, [Phys. Rev. Lett. **110**, 196403 \(2013\)](#).
41. J. L. M. van Mechelen, D. van der Marel, C. Grimaldi, A. B. Kuzmenko, N. P. Armitage, N. Reyren, H. Hagemann, and I. I. Mazin, Electron-phonon interaction and charge carrier mass enhancement in SrTiO₃, [Phys. Rev. Lett. **100**, 226403 \(2008\)](#).
42. A. Geondzhian, A. Sambri, G. M. De Luca, R. Di Capua, E. Di Gennaro, D. Betto, M. Rossi, Y. Y. Peng, R. Fumagalli, N. B. Brookes, L. Braicovich, K. Gilmore, G. Ghiringhelli, and M. Salluzzo, Large polarons as key quasiparticles in srtio₃ and srtio₃-based heterostructures, [Phys. Rev. Lett. **125**, 126401 \(2020\)](#).
43. C. Verdi, F. Caruso, and F. Giustino, Origin of the crossover from polarons to fermi liquids in transition metal oxides, [Nat. Commun. **8**, 15769 \(2017\)](#).
44. G. Whitfield and R. Puff, Weak-coupling theory of the polaron energy-momentum relation, [Phys. Rev. **139**, A338 \(1965\)](#).

45. O. Goulko, A. S. Mishchenko, L. Pollet, N. Prokof'ev, and B. Svistunov, Numerical analytic continuation: Answers to well-posed questions, [Phys. Rev. B **95**, 014102 \(2017\)](#).
46. S. X. Zhang, D. C. Kundaliya, W. Yu, S. Dhar, S. Y. Young, L. G. Salamanca-Riba, S. B. Ogale, R. D. Vispute, and T. Venkatesan, Niobium doped TiO₂: Intrinsic transparent metallic anatase versus highly resistive rutile phase, [J. Appl. Phys. **102**, 013701 \(2007\)](#).
47. V. C. Birschitzky, L. Leoni, M. Reticcioli, and C. Franchini, [Machine learning small polaron dynamics](#) (2024), [arXiv:2409.16179 \[cond-mat.mtrl-sci\]](#).
48. S. Fratini and S. Ciuchi, Dynamical mean-field theory of transport of small polarons, [Phys. Rev. Lett. **91**, 256403 \(2003\)](#).
49. E. Yagi, R. R. Hasiguti, and M. Aono, Electronic conduction above 4 K of slightly reduced oxygen-deficient rutile TiO_{2-x}, [Phys. Rev. B **54**, 7945 \(1996\)](#).
50. Y. Furubayashi, T. Hitosugi, Y. Yamamoto, K. Inaba, G. Kinoda, Y. Hirose, T. Shimada, and T. Hasegawa, A transparent metal: Nb-doped anatase TiO₂, [Appl. Phys. Lett. **86**, 252101 \(2005\)](#).
51. B.-S. Jeong, Y. Heo, D. Norton, A. Hebard, J. Budai, and Y. Park, Properties of anatase Co_xTi_{1-x}O₂ thin films epitaxially grown by reactive sputtering, [Thin Solid Films **488**, 194 \(2005\)](#).
52. Y. Kang, H. Peelaers, and C. G. Van de Walle, First-principles study of electron-phonon interactions and transport in anatase tio₂, [Phys. Rev. B **100**, 121113 \(2019\)](#).
53. X. Lin, C. W. Rischau, L. Buchauer, A. Jaoui, B. Fauqué, and K. Behnia, Metallicity without quasi-particles in room-temperature strontium titanate, [Appl. Phys. Lett. **2**, 41 \(2017\)](#).
54. M. Setvin, C. Franchini, X. Hao, M. Schmid, A. Janotti, M. Kaltak, C. G. Van de Walle, G. Kresse, and U. Diebold, Direct view at excess electrons in tio₂ rutile and anatase, [Phys. Rev. Lett. **113**, 086402 \(2014\)](#).
55. R. Levy, J. LeBlanc, and E. Gull, Implementation of the maximum entropy method for analytic continuation, [Comput. Phys. Commun. **215**, 149 \(2017\)](#).
56. J. J. Kas, J. J. Rehr, and L. Reining, Cumulant expansion of the retarded one-electron green function, [Phys. Rev. B **90**, 085112 \(2014\)](#).
57. J.-J. Zhou, J. Park, I. Timrov, A. Floris, M. Cococcioni, N. Marzari, and M. Bernardi, Ab initio electron-phonon interactions in correlated electron systems, [Phys. Rev. Lett. **127**, 126404 \(2021\)](#).

58. P. Mitrić, V. Janković, N. Vukmirović, and D. Tanasković, Cumulant expansion in the holstein model: Spectral functions and mobility, *Phys. Rev. B* **107**, 125165 (2023).
59. V. M. Stojanović, P. A. Bobbert, and M. A. J. Michels, Nonlocal electron-phonon coupling: Consequences for the nature of polaron states, *Phys. Rev. B* **69**, 144302 (2004).
60. W. Li, J. Ren, and Z. Shuai, A general charge transport picture for organic semiconductors with nonlocal electron-phonon couplings, *Nat. Commun.* **12**, 4260 (2021).

METHODS

First-principles calculations. Our DFT calculations use plane-wave kinetic energy cut-offs of 100 Ry for LiF, 85 Ry for SrTiO₃, 85 Ry for anatase TiO₂, and 60 Ry for rutile TiO₂, together with lattice parameters of 4.06 Å for LiF, 3.9 Å for SrTiO₃, $a = 3.78$ Å ($c/a = 2.51$) for anatase TiO₂, $a = 5.14$ Å ($c/a = 0.65$) for rutile TiO₂. We use norm-conserving pseudopotentials^{61,62} for LiF, SrTiO₃ and anatase TiO₂, and ultrasoft pseudopotentials^{63,64} for rutile TiO₂. The Wannier functions are constructed using the following $\mathbf{k}$ -point grids: $8 \times 8 \times 8$ for LiF, $8 \times 8 \times 8$ for SrTiO₃, $6 \times 6 \times 6$ for anatase TiO₂, and $4 \times 4 \times 6$ for rutile TiO₂. For anatase TiO₂, we use the Yambo⁶⁵ code to compute the G_0W_0 corrected band structure for the lowest 10 conduction bands. The G_0W_0 calculation includes 500 bands to compute the polarization function with the Bruneval-Gonze terminator⁶⁶. We verified that our G_0W_0 band structure of anatase TiO₂ is consistent with previous work⁶⁷. To obtain phonon dispersions and e -ph interactions on coarse grids, we conduct DFPT calculations with the following $\mathbf{q}$ -point grids: $8 \times 8 \times 8$ for LiF, $4 \times 4 \times 4$ for SrTiO₃, $6 \times 6 \times 6$ for anatase TiO₂, $4 \times 4 \times 6$ for rutile TiO₂. For rutile TiO₂, we employ DFT+ U calculations with PBEsol functional⁶⁴ and a Hubbard- U parameter of 4.5 eV and accordingly compute the e -ph interactions with DFPT+ U ⁵⁷. We include a Hubbard- U correction for rutile TiO₂ because its phonons from PBEsol show a soft mode. We use a Hubbard- U parameter of 4.5 eV taken from a previous first-principles study⁶⁸. We have checked that using $U = 3.9$ eV, obtained in Ref.⁵⁴ from ab initio constrained random phase approximation calculations, gives electronic band structure and phonon dispersions nearly identical to our $U = 4.5$ eV results. Our DFPT+ U phonon dispersions in rutile TiO₂ is in good agreement with two sets of experimental measurements^{69,70} (see Supplementary Section II. L). For SrTiO₃, we carry out calculations on the cubic phase and employ the SSCHA method⁷¹ to compute the phonon

dispersion at 200 K including anharmonic effects, thus removing the dynamical instability and imaginary phonons^{72,73}. We include the long-range Fröhlich interaction in all fine-grid e -ph matrix elements^{21,74}. The band structures for all materials studied in this work are provided in Supplementary Section II. F.

Numerical implementation of FEP-DMC. Our FEP-DMC algorithm models polarons using ultra-fine $\mathbf{k}$ - and $\mathbf{q}$ -point grids. In the first step, we construct a table for the electron energies and Kohn-Sham wave functions, phonon energies, and compressed e -ph matrices² by obtaining singular vectors $u_{mn}^\gamma(\mathbf{k})$ and $v_{mn}^\gamma(\mathbf{Q})$ on these dense $\mathbf{k}$ - and $\mathbf{q}$ -point grids. Owing to our e -ph compression technique, the memory requirement to store this table grows only linearly with system size. We use a modified version of the compressed e -ph interactions compared to our original approach²: while the original paper only decouples the electron and phonon momenta, here we employ singular value decomposition (SVD) to split the electron momentum and phonon-mode variables, including both the phonon momentum $\mathbf{q}$ and the mode index ν . This modified compression allows us to further speed up the FEP-DMC calculations because it bypasses the transformation from atomic Cartesian coordinates to the phonon eigenmode basis once the singular vectors $u_{mn}^\gamma(\mathbf{k})$ and $v_{mn}^\gamma(\mathbf{Q})$ are tabulated on the fine grids. Additional information about the FEP-DMC calculations is provided in Supplementary Section I.

DATA AVAILABILITY

The datasets for LiF generated and analyzed in this study is available in the Materials Cloud repository⁷⁵. Additional data and information are available upon reasonable request.

CODE AVAILABILITY

The PERTURBO code used to generate the data sets is an open source software, which can be downloaded at <https://perturbo-code.github.io/>. The first-principle electron-phonon diagrammatic Monte Carlo routines are available at <https://github.com/yaoluo/FEP-DMC>.

ADDITIONAL INFORMATION

Supplementary information is available at [insert URL].

Correspondence and requests for materials should be addressed to Marco Bernardi.

61. J. P. Perdew and Y. Wang, Accurate and simple analytic representation of the electron-gas correlation energy, [Phys. Rev. B **45**, 13244 \(1992\)](#).
62. D. R. Hamann, Optimized norm-conserving vanderbilt pseudopotentials, [Phys. Rev. B **88**, 085117 \(2013\)](#).
63. D. Vanderbilt, Soft self-consistent pseudopotentials in a generalized eigenvalue formalism, [Phys. Rev. B **41**, 7892 \(1990\)](#).
64. K. F. Garrity, J. W. Bennett, K. M. Rabe, and D. Vanderbilt, Pseudopotentials for high-throughput DFT calculations, [Comput. Mater. Sci. **81**, 446 \(2014\)](#).
65. D. Sangalli, A. Ferretti, H. Miranda, C. Attaccalite, I. Marri, E. Cannuccia, P. Melo, M. Marsili, F. Paleari, A. Marrazzo, G. Prandini, P. Bonfà, M. O. Atambo, F. Affinito, M. Palummo, A. Molina-Sánchez, C. Hogan, M. Grüning, D. Varsano, and A. Marini, Many-body perturbation theory calculations using the yambo code, [J. Phys. Condens. Matter **31**, 325902 \(2019\)](#).
66. F. Bruneval and X. Gonze, Accurate GW self-energies in a plane-wave basis using only a few empty states: Towards large systems, [Phys. Rev. B **78**, 085125 \(2008\)](#).
67. E. Baldini, L. Chiodo, A. Dominguez, M. Palummo, S. Moser, M. Yazdi-Rizi, G. Auböck, B. P. P. Mallett, H. Berger, A. Magrez, C. Bernhard, M. Grioni, A. Rubio, and M. Chergui, Strongly bound excitons in anatase tio2 single crystals and nanoparticles, [Nat. Commun. **8**, 13 \(2017\)](#).
68. E. German, R. Faccio, and A. W. Mombrú, A DFT + U study on structural, electronic, vibrational and thermodynamic properties of TiO₂ polymorphs and hydrogen titanate: tuning the Hubbard U-term, [J. Phys. Commun. **1**, 055006 \(2017\)](#).
69. J. G. Traylor, H. G. Smith, R. M. Nicklow, and M. K. Wilkinson, Lattice dynamics of rutile, [Phys. Rev. B **3**, 3457 \(1971\)](#).
70. B. Wehinger, A. Bosak, and P. T. Jochym, Soft phonon modes in rutile tio₂, [Phys. Rev. B **93**, 014303 \(2016\)](#).
71. L. Monacelli, R. Bianco, M. Cherubini, M. Calandra, I. Errea, and F. Mauri, The stochastic self-consistent harmonic approximation: calculating vibrational properties of materials with full quantum and anharmonic effects, [J. Phys. Condens. Matter **33**, 363001 \(2021\)](#).
72. J.-J. Zhou, O. Hellman, and M. Bernardi, Electron-phonon scattering in the presence of soft modes and electron mobility in SrTiO₃ perovskite from first principles, [Phys. Rev. Lett. **121**, 226603 \(2018\)](#).

- 73. C. Verdi, L. Ranalli, C. Franchini, and G. Kresse, Quantum paraelectricity and structural phase transitions in strontium titanate beyond density functional theory, [Phys. Rev. Mater. **7**, L030801 \(2023\)](#).
- 74. C. Verdi and F. Giustino, Fröhlich electron-phonon vertex from first principles, [Phys. Rev. Lett. **115**, 176401 \(2015\)](#).
- 75. Y. Luo, J. Park, and M. Bernardi, First-principles diagrammatic Monte Carlo for electron-phonon interactions and polarons, [Materials Cloud Archive **2025.70**, \(2025\)](#).